# KAUSTUBH DWIVEDI

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## PERSONAL STATEMENT

I am an engineer who has never lost the curiosity that first drew me to machines. That fascination has grown into a particular interest in aircraft structures and the challenge of making them lightweight without compromising their strength. My experience in composite design and my MSc at Cranfield University have allowed me to explore this interest through aircraft component development, composite manufacturing and impact testing. I enjoy using simulation alongside physical testing to understand how structures behave and make better design decisions.

## EDUCATION

**MSc, Advanced Lightweight and Composite Structures,** **September 2025 - September 2026**
**Cranfield University, Cranfield, UK**

- **Weighted average: 78.83%** | Relevant study: composite structures, finite element analysis, structural stability and impact mechanics
- **Thesis:** Slat Track Cans of the Future: Exploring the Design Space of a High Criticality Fuel-Tank Component

**B.Tech, Mechanical Engineering,** **July 2020 - July 2024**
**Symbiosis Institute of Technology, Pune, India**

- CGPA: 8.36/10

## ENGINEERING EXPERIENCE

**MSc Thesis: Slat Track Cans of the Future:**
**Exploring the Design Space of a High Criticality Fuel-Tank Component**
**GKN Aerospace** | MSc research project at Cranfield University **May 2026 - September 2026**
**Tools: CATIA V5, Abaqus, Tosca Structure, nCode and ANSYS Additive.**

- Redesigned an Airbus A320 slat-track can in AlSi10Mg, **reducing mass by 20.6%** while retaining elastic response across **five load cases**.
- In a separate variable-wall comparison, reduced simulated displacement and stress relative to the uniform-wall model while avoiding local yielding.
- Predicted **1.145 million fatigue repeats** for the existing welded assembly against a **300,000-repeat target** using nCode and EN 1999-1-3 curves; verified calculations independently within **3%**.
- Delivered a watertight CATIA V5 solid and STEP model, checked the design using an independent mesh and investigated build-direction thermal hot spots in ANSYS Additive.
- Screened four manufacturing routes and documented outstanding material-data, buckling and leak/proof-test requirements.

**Team Lead: Effect of Bird Strike on a Composite Wing Leading Edge**
**GKN Aerospace** | Group design project at Cranfield University **January 2026 - May 2026**

- **Led a four-person bird-strike project team**, allocating tasks, monitoring deadlines and coordinating simulation and testing activities.
- Ranked five leading-edge systems - CFRP, GFRP, Kevlar/epoxy, honeycomb sandwich and metal - at 90 and 180 m/s using **Abaqus/Explicit SPH-FEA**, comparing deformation, force response and energy absorption.
- Presented design-review recommendations, explaining laminate behaviour, manufacturing trade-offs and modelling limitations to guide material selection.
- Conducted impact tests on composite panels and honeycomb sandwich panels at Cranfield Impact Centre and investigated correlation with bird-strike simulations.

**Mohan Enviro Engineering Pvt Ltd, Ghaziabad, India** **December 2023 - April 2024**
**Mechanical Design Engineer**

- Designed fibre-reinforced plastic (FRP) tanks in CATIA V5, selecting wall thicknesses and laminates.
- Performed design calculations and stress analysis to assess structural performance.
- Prepared assembly drawings and maintained supporting design documentation.
- Produced bills of materials (BOMs) documenting material and component requirements.

**La Fondation Dassault Systemes / Dassault Systemes, Pune, India,** **December 2022 - December 2023**
**Engineering Design Intern**

- Developed a modular multi-ratio gearbox **from concept to working prototype** using **CATIA V5, 3DEXPERIENCE**, analytical calculations and **Abaqus**.
- Refined modular gearbox geometry against three design constraints: force transmission, packaging and manufacturability, updating technical documentation after each iteration.
- Collaborated with senior engineers during the placement, translating formal review feedback into gearbox changes and maintaining technical documentation through design sign-off.

## TECHNICAL SKILLS & PROFESSIONAL MEMBERSHIP

- **Software: CATIA V5 (including Composite Design workbench), 3DEXPERIENCE, Abaqus/Standard, Abaqus/Explicit, ANSYS Additive, nCode, Tosca Structure, Python, MATLAB and Excel.**
- **Structural Engineering:** nonlinear finite element analysis, contact modelling, SPH impact, mesh sensitivity and independent-model verification.
- **Composites and Testing:** laminate and sandwich design, stacking sequences, hand lay-up, curing, specimen manufacture, polymer compression testing, impact testing and experimental data interpretation.
- **Fatigue and Verification:** EN 1999-1-3 curves, CS 25.963 and CS 25.561 load definitions, welded-joint fatigue assessment and requirements verification matrices.
- **Data Analysis:** Applied particle swarm optimisation to two predictive models, Random Forest and XGBoost, to estimate lattice mechanical performance from experimental manufacturing parameters.
- **Additive Manufacturing:** wall-thickness design, build orientation, support strategy and manufacturing-risk assessment.
- **Professional membership:** Royal Aeronautical Society, Student Affiliate, since November 2025.

## PUBLICATIONS

- Designed, manufactured and compression-tested PLA+ diamond lattices to link printing parameters and cell geometry with mechanical performance, producing **first-author research** in Materials Today Communications, DOI 10.1016/j.mtcomm.2024.108233 (2024).
- Collaborated with co-authors on a published comparison of five aerospace material systems under bird-strike loading in the Journal of Manufacturing and Materials Processing, DOI 10.3390/jmmp10090343 (Cranfield University, 2026).
- Co-authored three peer-reviewed chapters covering polymer weld-strength optimisation and classification of dimensional deviation and surface roughness in 3D-printed parts, DOI 10.1201/9781003437369 (CRC Press, 2024).